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Poster: Thinking Through Musical Parameter Space

Abstract

The aesthetics of the musical output of a music generation system are often determined a small number of high-level parameters. Choosing a trajectory through this 'parameter space' is one of the main artistic decisions facing the designer of such automated music systems. Control of these parameters is often achieved by a musician by directly manipulating various knobs and sliders. The Mind-Modulated Music Interface (MMMI) is a system where control of these high-level parameters is provided in real-time by measuring the listener's brain with an EEG.

By monitoring a neural synchrony parameter which has been linked to 'musical processing', the MMMI provides the music generation engine with a time-varying indication of the listeners attention to the stimulus. This information can then be used in various ways to adapt and modulate the stimulus by mapping it to the appropriate aesthetic parameters. in this way, the listener is no longer a passive observer of the musical system, but actively involved in it's evolution and journey through the musical parameter space it occupies.

A working prototype of the MMMI has been constructed, and a detailed user study is in the planning stage. The authors goal is both to build a system for his own musical expression, and also to suggest a paradigm by which other artists can harness the power of the brain to experience and control their work in a new way.